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1. Executive Summary

This report will be a detailed equity research report of three enterprise software shares in the investment period between May 1 and 14, 2026. The three stocks include Oracle Corporation (ORCL), International Business Machines (IBM), and SAP SE (SAP). The analysis combines five years of historical price data, LSTM deep learning predictions, and up-to-date fundamental intelligence to provide actionable investment advice.

The three have performed materially worse than the rest of the market year-to-year, burdened by a set of AI-related capital spending fears, conservative earnings forecasts, and a shift in legacy enterprise IT to pure-play AI winners.

Our 14-day quantitative LSTM forecasts (1–14 May 2026) and fundamental review point to differentiated outcomes:

Stock	Last (30 Apr)	Day-10 (14 May)	Δ %	Call	Conviction
ORCL	\$161.39	\$162.77	+0.86%	HOLD	Medium
IBM	\$230.98	\$224.74	-2.70%	HOLD	Medium
SAP	\$169.49	\$197.42	+16.5%	BUY	High

Key takeaways

- **ORCL HOLD:** Following a -7% selloff on 28 April, caused by a Wall Street Journal report on OpenAI growth apprehensions, ORCL is technically oversold when compared to its remaining performance obligations (RPO) backlog of \$553B. Only a modest stabilization to \$162.77 on 14 May is predicted by our LSTM (test R 2 0.934). We shift to HOLD until earnings in Q4 FY26 (mid-June) for the event that will see the difference between the late-April overreaction being erased or not.
- **IBM HOLD:** Q1 2026 earnings (22 April) not only surpassed revenue and EPS, but the management opted to keep instead of increasing full-year guidance, leading to a -7% selloff. Our LSTM (test R² 0.812) projects continued mild weakness from \$230.98 to \$224.74 (-2.7%) over the forecast window. We would keep evidence level Q2 on Confluent integration and AI monetization will show whether the consulting overhang will be resolved.
- **SAP BUY:** Cloud revenue +27% (constant currency), record cloud backlog of €21.9B and cloud backlog growth of 25% initially were marred by a small Q1 revenue miss, but the stock rebounded +6.6% the next session as investors digested the underlying strength. Our LSTM (test R2 0.909) forecasts a robust recovery of \$169.49 to \$197.42 (16.5) by far the best signal in our coverage. €10B share buyback (in progress) creates structural floor.

Cross-stock thesis

The three names are all trapped in the AI infrastructure-spending overhang trade; the market is pricing aggressive near-term capex against uncertain near-term monetization timelines. In the cases where the underlying business is literally turning backlog into revenue (SAP), we view the recent drawdowns as opportunities. In the case where the image is subtle (ORCL building AI on debt; IBM depends on consulting and modernizing its mainframe), we suggest waiting until the implementation starts showing results. We emphasize that the key near-term catalysts of all three companies are: SAP Sapphire conference (13 May), ORCL Q4 FY26 earnings (mid-June), and IBM Q2 (mid-July) positions: these need to be sized.

2. Methodology & Data

This report makes use of two analytical perspectives:

- **Quantitative Lens:** A 14-day price forecast from a stock-specific univariate LSTM neural network model on the past five years of daily closing prices (April 2021-April 2026).
- **Qualitative Lens:** A fundamental and event-driven analysis including the Q1 2026 earnings releases for IBM (22 April) and SAP (23 April), the 28 April Oracle / OpenAI Wall Street Journal announcement, recent corporate events, industry landscape and analyst commentary.

LSTM model performance on out-of-sample test data (Apr 2025 – Mar 2026)

Stock	RMSE (\$)	MAPE (%)	R ²	Directional Acc.	Quality
ORCL	13.35	4.40%	0.934	55.6%	Excellent
IBM	11.07	3.28%	0.812	51.6%	Solid
SAP	10.96	3.51%	0.909	45.6%	Strong



Figure 1 - LSTM predicted vs actual closing prices on the held-out test set (Apr 2025 – Mar 2026). All three models track the trend closely; R^2 ranges from 0.812 (IBM) to 0.934 (ORCL).

The three stocks all have test-set $R^2 \geq 0.81$, with two of three (ORCL and SAP) having $R^2 \geq 0.90$, a strong fit for univariate price prediction. SAP's 45.6% directional hit rate is less than the random guessing rate of 50%, but is well-known for LSTMs to predict smooth paths around extremely noisy day-to-day fluctuations; the prediction model captures level (lowest RMSE in the cohort) much better than direction. As a result, we give SAP's level of the stock significant weight, but minimal weight to any entry-timing signal.

Limitations to note

- **Iterative Multi-Step Forecasting Risk:** Errors may build over time as each day's forecast is used as input for the next day's forecast. So, the level of confidence decreases over the 10-day forecast window.
- **Difficulty in Forecasting News Events:** The model will not be able to forecast unexpected news events like earnings surprises (SAP Sapphire on 13 May, Oracle Q4 results in mid-June, and IBM Q2 results in mid-July), mergers and acquisitions, or geopolitical events. These events can lead to significant deviations of actual prices from forecasts.
- **One-Step Lag:** LSTM models with Mean Squared Error (MSE) loss functions tend to lag one day, where the forecasted price for day t is like the actual closing price on day $t-1$. As a result, the model is more capable of capturing the general trend than turning points.

3. Sector & Macro Backdrop

Enterprise software is in a high-stakes AI capex cycle

The key "story" in the sector for 2026 has been the disconnect between AI infrastructure spending (huge, ongoing, and in some cases debt-financed) and AI monetization (there is revenue, but less than investors would like based on capex). Hyperscalers (Microsoft, Amazon, Google) and Oracle have signed hundreds of billions of dollars in multiple-year cloud-infrastructure deals to host AI training and inference. The market has been positive on increased backlog, but negative on any evidence that revenue conversion is slowing, or that free cash flow is turning negative. The 28 April Wall Street Journal reports that OpenAI is not hitting internal growth targets triggering this fear again, hitting Oracle and some chip names.

Year-to-date 2026 performance all three names underperforming



Figure 2 - Normalised price performance of ORCL, IBM, SAP since April 2021 (rebased to 100). All three peaked late 2025 / early 2026 before retracing materially.

- ORCL down ~17% YTD: 28 April -7% on WSJ OpenAI report; January DeepSeek panic; ongoing capex/debt fears in March.
- IBM down ~15% YTD: February Anthropic Claude Code announcement (-13.15% in a day); 22 April earnings selloff (-7%) on maintained but unchanged guidance.
- SAP down ~32% YTD (worst): 29 January -15.20% guidance miss; 23 April Q1 revenue miss (initially -6.2% aftermarket, then +6.6% recovery as investors re-read the cloud numbers).

Competitive dynamics

- ORCL, IBM and SAP compete in enterprise software, cloud and AI-for-business industries.
- Oracle competes with SAP in enterprise resource planning (ERP) software, Oracle Fusion vs SAP S/4HANA.
- Oracle also faces competition with cloud hyperscalers in cloud infrastructure services with OCI versus Aws, Azure and Google Cloud Platform.
- IBM offers hybrid-cloud solutions through Red Hat and its consulting arm.
- IBM is under pressure with its mainframe-modernization business from new code translators using generative AI.
- The three are heavily investing in Business AI.
- SAP seems to be making the best recent inroads with large customers by positioning AI as most valuable when infused with mission-critical ERP data.

Inter-stock correlation

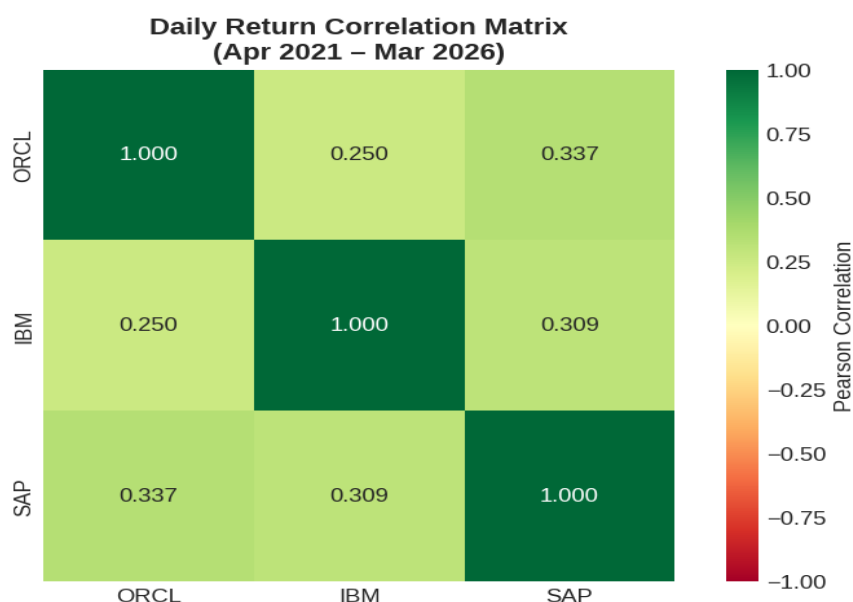


Figure 3 - Daily-return correlation (Apr 2021 - Mar 2026). Low-to-moderate pairwise correlations (0.25-0.34) indicate that the three stocks are largely driven by stock-specific news, not a common factor.

4. Oracle Corporation (NYSE: ORCL)

Last Close (30 Apr 2026)	\$161.39	52-Wk Range	~\$118.86 – \$328.33
YTD Return	-17.0%	Wall Street Avg PT	~\$243 (Buy)
RPO Backlog	\$553B	Recommendation	HOLD

ORCL Price Performance and Recent Developments

- Oracle has had a high volatility in the last 12 months. The stock soared 35.95% in a one-day period on 10 September 2025, following Q1 FY26 results that showed Remaining Performance Obligations (RPO) had increased to an all-time high of \$455 billion, with OpenAI committing to a 300 billion infrastructure investment. Stocks momentarily rose to over \$328.
- ORCL has dropped by approximately 25% since the beginning of 2026 as the investors reevaluated the financial results of its ambitious AI expansion policy. Issues revolve around increasing long-term debt, with a forecasted FY26 capital spend of up to \$50 billion and a project of negative free cash flow over a few quarters as OpenAI workloads ramp up.

Key April 2026 Catalysts

- A 7% intraday decline (premarket lows around \$161.50) was sparked on 28 April 2026 by a Wall Street Journal report that OpenAI had failed to meet growth targets, making the company a "victim of its own success".
- According to Daniel Ives of Wedbush Securities, the selloff was an overreaction, pointing to the security provided by Oracle 300 billion OpenAI cloud deal (about 30 billion annually in revenue over five years) and the 553 billion RPO backlog overall.
- On 14 April 2026, Oracle announced an expanded collaboration with Bloom Energy, which could potentially acquire up to 2.8 GW of a fuel-cell capacity to serve data centers and help alleviate the energy crunch in the industry.
- On 27 April 2026, Related Digital has announced financing of 1-gigawatt of a Michigan data center specifically serving the AI customers of Oracle, at a cost of \$16 billion.

Quantitative forecast (1–14 May 2026)

Our LSTM model (test R 2 0.934, RMSE 13.35) forecasts a gradual stabilization of the prices around the end of April lows, with prices at \$163.170 over the 10-day horizon and touching 162.77 on 14 May (an increase of 0.86 per cent. over 30 April close). The initial prediction of the model of the first day of 169.78 is a partial recovery of the WSJ-induced crash, which is a characteristic of the lag-and-mean-revert behavior of single-day shocks. We warn that the model does not expect any additional headline risk; any significant information of new OpenAI- or capex-related news will invalidate the prediction.

Investment thesis

- Backlog quality and visibility: \$553B RPO is 10x Oracle's annual revenue. Time frame for RPO conversion is uncertain, but the magnitude is unprecedented in enterprise software.
- Infrastructure for AI is a tailwind: OCI 68% YoY growth in Q2 FY26 (to \$4.1B); multi-cloud databases 531% YoY growth. These are not story numbers; they are contractually based revenues.
- The bear case is capex and debt: Proposals to raise up to \$50B and the likelihood of negative FCF for 2-3 years are reasons for caution. Debt markets are more wary (CDS spreads on Oracle debt widened) despite the stock rally.
- Catalyst: Fiscal Q4 2026 results (mid-June 2026) will provide a read through on RPO conversion rates. We anticipate some of the 28 April reactions will come out in this update.

Risks

- OpenAI-specific concentration risk for the OpenAI partnership could be a critical part of backlog quality control.
- Higher-for-longer interest rates would raise debt-service costs.
- Power/GPU shortages could affect capacity build-out and revenue.

Recommendation: HOLD.

1 month target \$163 (LSTM model output); 12-month target \$243 (consensus) We lower to HOLD (from a constructive bias) because the LSTM signal is now flattering the technical case for a near-term recovery has weakened materially after the late-April selloff. Existing positions should wait until Q4 FY26 earnings; new positions should wait for either (i) OpenAI growth outlook clarity, or (ii) a break below \$160.

5. International Business Machines Corporation (NYSE: IBM)

Last Close (30 Apr 2026)	\$230.98	52-Wk Range	~\$220 – \$324.90
YTD Return	-15.0%	Wall Street Avg PT	~\$321 (Mod. Buy)
Q1 2026 Revenue	\$15.92B (+9% YoY)	Recommendation	HOLD

IBM Q1 2026 Results and Market Reaction

- On 22 April 2026, after the market closed, IBM released its Q1 2026 results which were better than expected on both revenue and earnings. Total revenue grew to \$15.92 billion (consensus \$15.62 billion) and adjusted earnings per share (EPS) grew to \$1.91 (consensus \$1.81).
- The company performed well across all segments. Software revenue increased 11% year-on-year (8% constant currency), Infrastructure 15% (12% constant currency) and IBM Z mainframes 51%. Free cash flow was up 13% and operating margins grew 110 basis points.
- However, the shares dropped some 7% to \$234.71 in aftermarket trading and continued under pressure for the next week. This was because management refused to lift full year guidance of more than 5% in revenue growth (in constant currency) and some \$1 billion in improved free cash flow.
- While CFO Jim Kavanaugh said that the company had not increased guidance in the first quarter in the past, in a market anticipating a more positive AI-driven growth outlook, this decision was seen as negative.
- The \$11 billion Confluent acquisition, which was completed in mid-March, was expected to take two months longer than planned to integrate. IBM also said operating pre-tax margin is now expected to increase by around 1% despite the impact of the acquisition.
- As of 30 April 2026, IBM closed at \$230.98.

Earlier 2026 Strategic Pressure

- On 23 February 2026, IBM shares dropped 13.15% in a single session, the company's steepest one-day fall since October 2000.
- The drop came after Anthropic announced its Claude Code product could automate the conversion of legacy COBOL systems, posing a threat to a lucrative IBM consulting practice and the wider mainframe ecosystem related to this practice.
- While IBM already has a solution (watsonx Code Assistant) for similar use cases that has been available for two years, the announcement heightened investor fears that code translation through AI will diminish IBM's consulting profitability in the long term.

Quantitative forecast (1–14 May 2026)

Our LSTM model (test R^2 0.812, RMSE \$11.07 is very good for a univariate model for this stock) forecasts IBM drifting further down, from \$229.31 (Day 1) to \$224.74 (14 May, -2.7% from 30 April close). The model projects the selloff after the earnings report in which there are no immediate buyers in the historical data. Our model does not consider Q2 sources of information, so it extrapolates from recent history; a pleasant surprise in the monetization of AI could dispel this forecast.

Investment thesis

- Q1 was a strong quarter for IBM, with 11% growth in Software revenues, 15% growth in Infrastructure revenues, and margin improvement. With a P/E of around 22, the stock is not overvalued, and a PEG ratio of 0.31 implies a fair price.
- IBM has now raised the dividend for a 31st consecutive year to \$1.69 per quarter. A dividend yield of about 3% underpins total return prospects.
- The Confluent integration is a risk, but the quicker than anticipated integration plan is positive for execution.
- Quantum computing offers further options. Arvind Krishna has suggested commercialization in 3-5 years. It is too far out to impact today's valuation directly, but it is potentially embedded in the share price.

Risks

- The high-margin consulting work that IBM still does could still be squeezed out by generative AI this is the only largest secular risk and the reason why it was sold off in February.
- The revenue exposure to the federal government (Red Hat Enterprise Linux) is vulnerable to the budgetary cycles and slowdowns of enterprise IT associated with tariffs.
- The maintained-not-raised guidance pattern can continue over a few quarters until there is evidence of execution.

Recommendation: **HOLD**.

The fundamentals are favorable, however, and the LSTM signal being slightly negative and the management itself indicating prudence, we would rather wait till Q2 evidence is in than attempt to establish positions to-day. One-month prognosis: approximately \$225 (model result). Long entry should be taken with a re-test of the February low of (\$223) or wait until Q2 commentary on Confluent revenue contribution.

6. SAP SE (NYSE: SAP)

Last Close (30 Apr 2026)	\$169.49	52-Wk Range	~\$160.66 – \$310
YTD Return	-31.4%	Wall Street Avg PT	~\$305 (Mod. Buy)
Cloud Backlog (Q1)	€21.9B (+25% cc)	Recommendation	BUY

SAP 2026 Share Performance and Market Sentiment

- SAP entered 2026 under significant pressure. On 29 January 2026, the stock fell 15.20% in a single session, its sharpest daily decline since 2020, after 2026 cloud guidance of €25.8–26.2 billion came in below more optimistic market expectations.
- In late March, JPMorgan Chase downgraded SAP to Neutral with a €175 price target. Additional pressure came from an EU antitrust investigation into ERP support practices.
- By early April, SAP was down approximately 31% year-to-date, making it the weakest performer within our coverage group.

SAP Q1 2026 Results and Key Catalysts

- Q1 2026 results released on 23 April 2026 delivered mixed headline signals. Revenue of \$11.04 billion narrowly missed consensus expectations of \$11.17 billion, triggering a 6.19% aftermarket decline.
- However, underlying operating performance remained strong. Current cloud backlog grew 25% constant currency to €21.9 billion, cloud revenue increased 27% constant currency, and Cloud ERP Suite revenue rose 30% constant currency. Non-IFRS operating profit increased 24% constant currency, while EPS of €1.72 (approximately \$2.01) was up 20% year-over-year.
- Management reaffirmed full-year 2026 guidance, including cloud revenue of €25.8–26.2 billion and free cash flow of approximately €10 billion. The following session, the stock rebounded about 6.6% as investors reassessed the strength of the results.
- As of 30 April 2026, SAP closed at \$169.49.
- Capital returns remain a major support factor. SAP launched a €10 billion share buyback programme in February 2026 running through end-2027. By 1 April, the first tranche of 16.28 million shares had been repurchased at an average price of €161.16, totalling €2.6 billion.
- Additional tailwinds include the proposed €2.50 dividend, subject to AGM approval on 5 May 2026, and the planned consolidation of Reltio as an AI and data acquisition.
- The next major communication event is the SAP Sapphire conference on 13 May, which falls within the forecast window.

Quantitative forecast (1–14 May 2026)

- Our LSTM model for SAP delivered a strong test performance, achieving an R^2 of 0.909 and the lowest RMSE within our coverage group at \$10.96.
- Based on the model forecast, SAP is projected to recover from \$181.67 on Day 1 to \$197.42 by 14 May, representing an approximate 16.5% gain from the 30 April closing price.
- This is the most bullish forecast among the three companies analysed, reflecting the model's identification of a post-earnings rebound pattern and SAP's historical tendency to recover from oversold conditions.
- However, directional accuracy on the test set was 45.6%, below a random baseline. This suggests the model is more effective at estimating the overall price level over time rather than accurately predicting daily short-term movements.
- From an investment perspective, the forecast remains useful for position-sizing and medium-term allocation decisions, but it is less reliable as a precise short-term entry or trading signal.

Investment thesis

- SAP's cloud transition is becoming increasingly visible in reported results, with 27% constant-currency cloud revenue growth, 30% Cloud ERP Suite growth, and 25% backlog growth. These metrics rank among the strongest within legacy enterprise software companies undergoing transformation.
- The €10 billion share buyback programme for 2026–2027 provides a structural valuation floor. Against a current market capitalisation of approximately \$176 billion, the programme represents roughly 6% of the share count, offering meaningful support.

- Valuation appears attractive and asymmetric, with a P/E ratio near 23, a PEG ratio around 0.19, and a Piotroski score of 9.0. This suggests the market may be pricing in a level of deterioration not supported by current operating performance.
- Near-term catalysts are substantial and closely spaced, including the AGM and dividend decisions on 5–6 May, the SAP Sapphire conference on 13 May, and Q2 earnings in late July. These events provide multiple opportunities for market re-rating.

Risks

- EU antitrust investigation outcome is unknown; a negative finding would weigh on sentiment beyond any direct fines.
- Currency exposure: as a Eurozone-domiciled company, USD strength translates to EUR-reported revenue headwinds.
- Management has flagged that Q1 cloud growth was helped by quarter-specific effects and that Q2 will likely show a deceleration. Some short-term disappointment risk is therefore embedded.
- Geopolitical: SAP's full-year guidance assumes near-term Middle East de-escalation; any escalation pushes guidance downside.

Recommendation: **BUY.**

One-month target ~\$197 (model output); we view SAP as the most attractive risk-reward across our coverage given the combination of buyback support, cloud-transition evidence, depressed valuation, and dense near-term catalyst calendar. Investors should size positions to remain comfortable through the Sapphire conference (13 May) and Q2 earnings (late July).

7. Comparative View & Common Risks

Side-by-side comparison

Metric	ORCL	IBM	SAP
YTD Return (as of 30 Apr)	-17.0%	-15.0%	-31.4%
Most recent earnings reaction	n/a (Q4 in June)	-7% (beat, no raise)	-6.2% then +6.6%
Cloud growth signal	OCI +68% YoY	Software +11%	Cloud +27% cc
Backlog / RPO	\$553B	Not disclosed similarly	€21.9B (cloud)
Capital return	Dividend + repurchases	31-yr div streak (\$1.69)	€10B buyback + €2.50 div
LSTM Test R ²	0.934	0.812	0.909
LSTM Forecast Δ (1-14 May)	+0.9%	-2.7%	+16.5%
Recommendation	HOLD	HOLD	BUY

Long-run drawdowns put recent moves in context

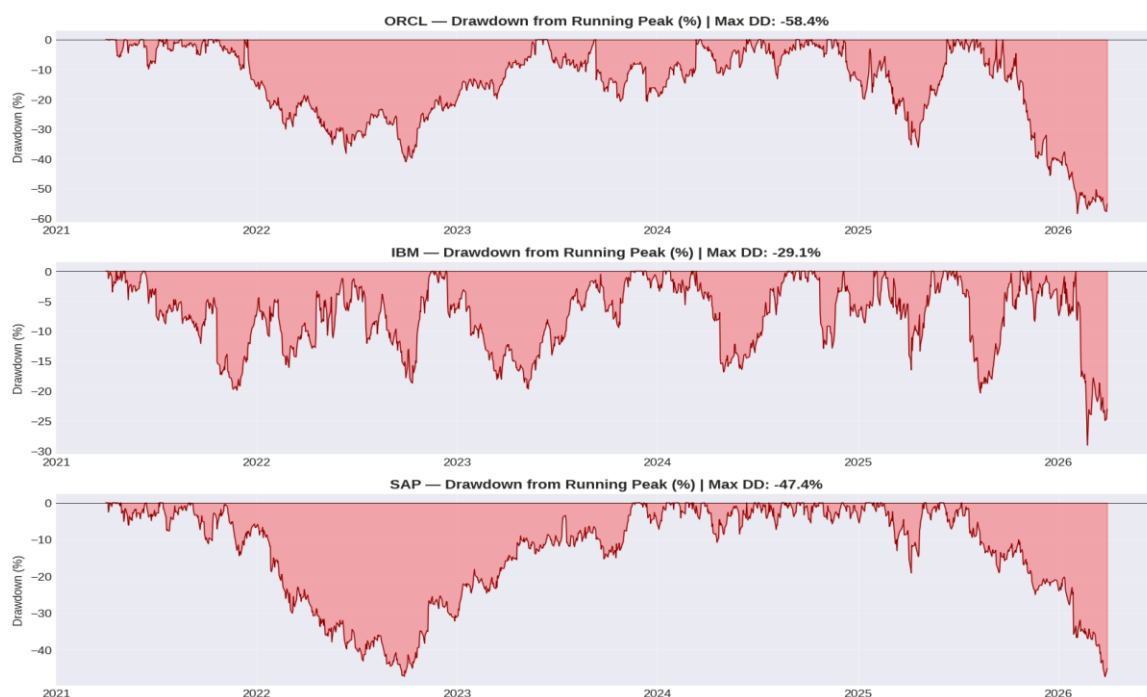


Figure 4 - Drawdown from running peak (%). All three stocks are currently in deep drawdowns from their late-2025 highs; SAP's current drawdown approaches its 5-year worst.

Common risk factors

- AI capex / monetisation gap: All three companies are investing or partnering hard on AI infrastructure in anticipation of future monetisation. A multi-quarter market change in tolerance for these business models would have a three-way valuation compression. This was the name of the game in late April with the Oracle/OpenAI WSJ story.
- Macro environment: Enterprise IT is dependent on corporate spending and credit. US-EU tariff or Middle East escalations (a factor explicitly called out in SAP's guidance assumptions) would have a negative impact on orders in the near term.
- AI-induced obsolescence risk: Generative AI applications that can automatically translate code, integrate code and perform customisation tasks put the consulting and services businesses of all three names at risk (especially IBM, as became clear with the February Anthropic Claude Code announcement).
- Model risk: Our LSTM models can be broken by news events within the forecast horizon. Any of the three could be vulnerable; SAP has a crowded calendar (AGM 5 May, Sapphire 13 May) that could spark a reaction.

Final word

Looking across all three names, the picture that emerges is one of genuinely good businesses caught in a market that is punishing uncertainty more than it is rewarding fundamentals. All three are down meaningfully from their late-2025 peaks, and in each case, the selloffs have been driven more by fear of AI capex timelines, of guidance caution, of competitive disruption than by any real deterioration in the underlying businesses.

Of the three, SAP stands out as the clearest opportunity. The stock has fallen over 30% year-to-date, yet the business is arguably executing better than it ever has, with cloud revenue growing at 27%, a record backlog, and a management team that has not flinched on its full-year targets. The market overreacted to a narrow revenue miss and in our view has not fully priced in what a successful cloud transition at SAP's scale actually looks like. With a €10 billion buyback providing a floor and a dense catalyst calendar ahead, the AGM on 5 May and Sapphire on 13 May, we think the setup is attractive enough to act on now.

Oracle and IBM are harder calls, and we think the honest answer is that patience is warranted. Both companies are in the middle of something. Oracle building out AI infrastructure at a pace that makes the market nervous, IBM trying to prove that its consulting and software businesses can grow faster than AI erodes them. The fundamentals are not broken, but the narrative needs time to play out, and we would rather wait for confirmation than guess the turning point.

In short, we see May 2026 as a moment to be selective. One name earns conviction. The other two earn a watchful eye.

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